

Metric-definition audit & threshold-provenance correction

Preflight for EVAL-NATQ2-H-001 · verdict **NOT_EQUIVALENT** · 0 runs consumed · reserve 0 bytes · head 195b073 · 2026-09-05 17:59Z

The historical 20/40 is not a NATQ-002 baseline and must stop being read as one. NATQ-001 `strict_recall@10` and NATQ-002 `case_hit@10` are **NOT_EQUIVALENT**. They differ on two independent axes at once — one counts a case only when every gold span is retrieved, the other when *any* one is; and one additionally requires the retrieved chunk to carry the same `section_path` as the gold span, which NATQ-002 gold evidence does not even record. On identical retrieval, `strict_recall@10 ≤ case_hit@10` always, so 0.50 is a *lower bound* on what the old number would have been under the new metric, not an estimate of it. The two are also measured on different question sets with no shared cases, so no paired reading survives even if the definitions were aligned.

NOT_EQUIVALENT
METRIC EQUIVALENCE

3/7
SYNTHETIC CASES THAT
SEPARATE THEM

yes
H AND BM25 SHARE A
DEFINITION

14/14
FROZEN ARTIFACTS
UNCHANGED

1 — Where the two metrics part company

Axis	NATQ-001 <code>strict_recall@10</code>	NATQ-002 <code>case_hit@10</code>	Status
Aggregation	ALL gold spans (fully_recalled)	ANY gold span	differs
Match predicate	<code>version_id</code> + <code>section_path</code> + char overlap	<code>version_id</code> + char overlap	differs
Evaluation depth	TOP_K = 10	top 10	same
Duplicate covering chunks	min(rank) over matches	first match in rank order	same
Tie handling	pipeline rank order	round(score,9) DESC, <code>chunk_id</code> ASC	deterministic in both
Case denominator	cases in partition (40)	cases in partition (40)	same

Read from the scoring code, not from the metric names. The ALL/ANY split lives in `score_system`'s `fully_recalled = found == len(spans)`; the predicate split lives in `dict_overlaps`, which tests `list(section_path)` equality that the NATQ-002 runner never tests. Sixteen of the forty validation cases carry more than one gold span, so the aggregation choice is live on 40% of the partition; `section_path` is missing from 100% of NATQ-002 gold evidence, so the predicate choice is not a tuning knob but a schema fact.

2 — Synthetic demonstration

Hand-constructed fixtures only. No corpus, no database, no benchmark case, no protected partition. The point is to show the divergence is a property of the definitions rather than of any dataset.

Fixture	strict@10	case_hit@10	full_cov@10
SYN-1 multi-span, one span found Aggregation differs: ANY-span passes, ALL-spans fails.	—	HIT	—
SYN-2 single span, chunk filed under a different section_path Match predicate differs: NATQ-001 requires section_path equality, NATQ-002 does not.	—	HIT	HIT
SYN-3 single span, exact section_path, overlapping chunk Control: the two definitions agree when the case is single-span and the paths match.	HIT	HIT	HIT
SYN-4 covering chunk returned below the evaluation depth Control: depth 10 is enforced identically by both.	—	—	—
SYN-5 two duplicate covering chunks at ranks 3 and 7 Control: duplicates collapse to the best rank in both; no double counting.	HIT	HIT	HIT
SYN-6 cross-document case, only the first document's span found Multi-document aggregation: each span resolves in its own version_id under both.	—	HIT	—
SYN-7 one-character overlap at the span boundary Both predicates are half-open interval intersection, so a single shared character hits.	HIT	HIT	HIT

SYN-1 multi-span, one span found, SYN-2 single span, chunk filed under a different section_path, SYN-6 cross-document case, only the first document's span found separate strict from case_hit. SYN-2 single span, chunk filed under a different section_path is the one that matters most: it still disagrees after the aggregation is matched, which is why case_full_coverage@10 is the closest analogue of the historical metric and still not the same function.

3 — SYSTEM-H and the frozen BM25 comparator

Same definition: True. The EVAL-NATQ2-H-001 span_hit clause and the frozen BM25 runner implement the same predicate at the same depth with the same aggregation and the same denominators, over the same 40 validation cases. The comparator's case_hit@10 = 0.375 is measured on exactly the metric the preregistration names.

Residual requirement. The SYSTEM-H runner, when written, must implement metrics.span_hit exactly as the BM25 runner does — document identity plus character overlap, no section_path term. A SYSTEM-H runner that reused the NATQ-001 overlaps()/dict_overlaps() helpers would silently score a stricter metric and would not be comparable to the frozen 0.375.

Recommended control. The SYSTEM-H runner should recompute the BM25 comparator's case_hit@10 from the frozen ranked output and refuse to proceed unless it reproduces 0.375, proving both systems went through one scorer.

4 — Threshold provenance: the contradictory row

Classification **PRESENTATION_ONLY** · affects the protocol: **no**.

Item	Value
As rendered in STATUS-V2-007	Empirically derived from NATQ-002 True (i.e. not derived)
Underlying field	<code>not_empirically_derived_from_NATQ_002 = true</code>
Which reads as	It is TRUE that the floors were NOT empirically derived from NATQ-002.
Corrected wording	Derived from NATQ-002 measurements — No — engineering qualification floors, set before any NATQ-002 score existed
Boolean flipped on assumption	no

The row label asserts 'empirically derived' and the value asserts 'True', which together state the opposite of the truth. The trailing parenthetical contradicts the row it is attached to rather than repairing it. The defect is that a negated field name (`not_empirically_derived_from_NATQ_002`) was rendered against a positively-worded label without inverting the value. The stored data was already correct in every artifact that governs the run, so nothing was amended: the floors remain 0.80 and 0.65 exactly as approved, and every hash is untouched. This is a label defect in one delivered table, recorded rather than quietly rebuilt.

5 — Are the qualification boundaries unambiguous?

- At 0.80: PASS is ≥ 0.80 , so exactly 0.80 qualifies.
- At 0.65: FAIL is < 0.65 , so exactly 0.65 does not fail.
- Between them: $0.65 \leq \text{case_hit@10} < 0.80$ with a significant positive delta is INCONCLUSIVE, stated explicitly in the preregistration.
- With 40 validation cases, `case_hit@10` moves in steps of 0.025; $0.65 = 26/40$ and $0.80 = 32/40$ are both exactly attainable, so neither boundary is unreachable.

One gap, flagged and not repaired: PROPOSED_PROTOCOL_CORRECTION — NOT APPLIED, REQUIRES

COORDINATOR APPROVAL. A significant REGRESSION falls through to INCONCLUSIVE. If SYSTEM-H's micro-MRR is worse than BM25 with a 95% interval excluding 0, and `case_hit@10` is 0.65 or above, then PASS is false (delta is not > 0) and FAIL is false (the interval does not include 0, and the `case_hit` floor is met), so the run is labelled INCONCLUSIVE. A system that is decisively worse than the lexical control would not be recorded as a failure.

Worked example — $\text{delta} = -0.06$, 95% interval $[-0.10, -0.02]$, `case_hit@10` = 0.70 -> INCONCLUSIVE under the rule as written.

Proposed replacement clause: *FAIL: the 95% interval includes 0, OR the 95% interval excludes 0 with a negative mean delta, OR `case_hit@10` < 0.65 .*

The run remains valid and interpretable; a decisive regression would simply be reported as INCONCLUSIVE with the negative delta visible in the results. This is a labelling weakness, not a source of bias, and it does not by itself block the run. Applying this would be a protocol change and must be issued as a replacement preregistration with its own hash, on an explicit coordinator ruling. It has not been applied here.

6 — What this changes about the run

- Definition-level: NOT_EQUIVALENT. `strict_recall@10` is strictly stricter than `case_hit@10` on both axes, so on identical retrieval `strict_recall@10` $\leq$ `case_hit@10` always. The historical $20/40 = 0.50$ is therefore a LOWER bound on what `case_hit@10` would have been on that dataset, not an estimate of it.
- Schema-level: the NATQ-001 predicate is not computable on NATQ-002 as it stands, because NATQ-002 gold evidence carries no `section_path`. Making the definitions equivalent would require adding a field to frozen gold evidence, which is prohibited and would in any case void the freeze.

- Dataset-level: even if the definitions were made identical, NATQ-001 and NATQ-002 are different question sets over the same corpus snapshot with no shared cases. A comparison across them is unpaired and carries no paired statistical interpretation.
- The preregistered decision rule does not depend on this. EVAL-NATQ2-H-001 tests SYSTEM-H against the co-run BM25 comparator on the same 40 NATQ-002 cases; the NATQ-001 prior enters only as disclosed context.

7 — New artifacts

File	SHA256
METRIC-AUDIT-001.json	5d18f071a83513b686e049e8c322100af454bf35e80c87ae0e9b8af096d03f39
METRIC-AUDIT-001-PROBE-OUTPUT.json	8177ab746590013d86f76706e10db4636a94da4ea514a3122fe36c71b5993f40
metric_equivalence_probe.py	4a985d44cb876836c6c738149d521bc803b5921f7d9c9bbd9b252ade15453672
THRESHOLD-PROVENANCE-CORRECTION-001.json	439e476879bb8575ad23ae2735e8a768057ab5f332b029c572842aca83dfbe35

Authoritative SYSTEM-H config hash `7599eb3c3bb4798230a722e6a7c96b046dc36cc0332b584ec55339896b2d717a` , preserved. No system was run, no run was consumed, no reserve or holdout was opened, and the BM25 comparator was neither rerun nor tuned. Prior status recorded at head `f492f07` .